Khmer Automatic Speech Recognition

Three student-adapted pretrained models on FLEURS Khmer

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Problem: turning Khmer audio into usable text

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People need Khmer transcripts for captions, interviews, and notes, but producing them by hand is slow.

> **Why is it difficult?**
>
> A transcriber repeatedly listens to each utterance, types Khmer text, and checks wording and spelling.

Project question: which adapted ASR model makes fewer character and word errors?

Data: FLEURS Khmer utterances

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One utterance = one short audio recording with a reference transcript.

531

training utterances

Whisper and MMS

124

validation utterances

checkpoint selection

114

held-out test utterances

all three models

Qwen training: 16,720 effective examples, including repeats and additional public speech.

Evaluation: CER and Khmer WER

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Character error rate (CER)

CER = (S + D + I) / N_chars × 100%

N_chars = reference characters after Unicode normalization and space removal.

Word error rate (WER)

WER = (S + D + I) / N_words × 100%

N_words = reference words from ICU Khmer word segmentation.

S = substitutions D = deletions I = insertions. Sum edits over all utterances before dividing.

Lower is better. WER can exceed 100% when the model inserts extra words.

Three approaches, all trained from existing models

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I started with pretrained checkpoints from Hugging Face, then adapted each one to Khmer ASR.

1 Whisper-Tiny

Starting model: openai/whisper-tiny

Full fine-tuning: updated all model weights

531 clips · 3 epochs
LR 1e-5

2 MMS-1B CTC

Starting model: facebook/mms-1b-all

Partial fine-tuning: top 4 encoder layers and CTC head

531 clips · 15 epochs
LR 3e-5 · best step 300

3 Qwen3-ASR 0.6B

Starting model: seanghay Khmer checkpoint

LoRA fine-tuning: trained a small adapter

16,720 effective · 3 epochs
LR 1e-5 · best epoch 1

Why these three models?

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Whisper-Tiny

Small encoder-decoder baseline; tests full fine-tuning of a widely used generative ASR model.

MMS-1B CTC

Large multilingual speech encoder; tests partial fine-tuning with CTC decoding.

Qwen3-ASR

Khmer-adapted generative model; tests parameter-efficient LoRA with more public data.

The choice compares model families and adaptation methods. Unequal training data prevents a causal architecture ranking.

How I trained and selected the models

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Whisper-Tiny

All weights on 531 clips. 3 epochs, Tesla T4.
8.7 min measured for the complete run.

MMS-1B CTC

Top 4 encoder layers and CTC head on 531 clips.
15 epochs, T4; about 1.6 h estimated.

Qwen3-ASR LoRA

LoRA on 16,720 effective entries. 3 epochs, T4.
About 6.3 h estimated active training time.

Workflow: prepare audio, fine-tune, select using 124 validation clips, then score once on the shared 114-clip test.

Results: shared held-out test

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Same 114 FLEURS Khmer test utterances for all three. LR = learning rate; lower error is better.

Approach	Training examples	Training setup	CER (%)	ICU WER (%)
Whisper-Tiny	531	3 epochs LR 1e-5	86.76	106.27
MMS-1B CTC	531	15 epochs LR 3e-5	15.84	67.16
Qwen3-ASR LoRA	**16,720 effective**	3 epochs LR 1e-5	**13.71**	**33.40**

Qwen leads on this test, but its larger training set and Khmer-adapted base prevent a controlled comparison.

Qwen checkpoint selection and final test

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Validation · 124 utterances

12.42	27.99
CER (%)	ICU WER (%)

Held-out test · 114 utterances

13.71	33.40
CER (%)	ICU WER (%)

Epoch 1 was chosen by validation WER before the one-time test. Its 27.99% validation WER missed the <25% goal.

Critical analysis of the test results

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Whisper-Tiny

2,644 word edits for 2,488 reference words. Its 106.27% WER shows this checkpoint is unreliable on this test.

MMS-1B

15.84% CER but 67.16% WER. Character-level improvement does not mean whole-word accuracy; 605 of 1,671 word edits were insertions.

Qwen3-ASR

831 word edits, 840 fewer than MMS. The larger corpus and Khmer-adapted base may contribute; this test cannot separate their effects.

Limits and next analysis

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Mixed-script text

MMS CER: 23.75% on 26 references with Latin letters versus 13.03% on 88 others.

Data imbalance

Qwen used 31.5× as many effective training examples and a Khmer-adapted base.

Test coverage

114 FLEURS utterances do not represent every speaker, microphone, or topic.

Next: listen to difficult utterances; then rerun a matched-data comparison.

Conclusion

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Qwen3-ASR had the lowest observed test error

13.71% CER **33.40% ICU WER**

The <25% validation WER goal remains unmet. Qwen had more training data, so the result does not establish an architecture advantage.

Thank you. Questions?

References and project evidence

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1. FLEURS: Conneau et al., Few-shot Learning Evaluation of Universal Representations of Speech.
2. Whisper: Radford et al., Robust Speech Recognition via Large-Scale Weak Supervision.
3. MMS: Pratap et al., Scaling Speech Technology to 1,000+ Languages.
4. Qwen Khmer base: seanghay/Qwen3-ASR-0.6B-Khmer on Hugging Face.
5. Evaluation: ICU Khmer word breaks and saved project prediction pairs.

Full run logs and prediction pairs are saved in the project repository.